Efficient Global Content Recovery for Multi-View Data

Ryan Jackson, David Zhao, Dowoo Kim

1 Introduction and Motivation

In various real-world machine learning contexts, a single underlying environment is observed through multiple views, often spanning different modalities. These distinct views are generated by a complex mixture of hidden causal factors. Some of these latent factors are shared across the views (the "content"), while others are entirely specific to a single modality (the "style"). Recovering this shared content is a primary goal of representation learning, as it allows us to discard irrelevant, view-specific noise and isolate the robust, causal variables that define the true state of the environment.

Recently, Yao et al. introduced a comprehensive framework in *Multi-View Causal Representation Learning with Partial Observability* [1]. Their work establishes two fundamental theoretical guarantees. First, they prove that the shared content of a single set of views can be *block-identified*, provided the exact dimensionality of this shared content is known prior to training. Second, they demonstrate that by introducing a learnable masking mechanism and evaluating the shared content across all possible subsets of views simultaneously, one can identify the shared content of any subset without the need to retrain encoders for each specific combination. Crucially, this masking approach also overcomes the restrictive assumption of needing to know the exact content dimension beforehand.

While theoretically powerful, evaluating all possible view subsets simultaneously becomes computationally intensive as the number of views increases. This complexity is often unnecessary in practice, where interest is typically restricted to the global content—the latent variables shared across all views—rather than every possible intersection. Furthermore, while the masking approach of Yao et al. [1] addresses the challenge of an unknown content dimension, their framework assumes that the dimension of the latent space generating each individual view is known, which is rarely the case in real-world applications. Finally, their proposed information-sharing regularizer used to learn the masks relies on an L_0 penalty, which is highly non-convex and challenging to optimize in practice.

As such, in section 4, we propose solutions to these challenges by (i) restricting the focus to global content recovery to eliminate combinatorial overhead, (ii) relaxing the requirement for known view-specific dimensions by only requiring overestimation of the content dimension, and (iii) utilizing the differentiable hard concrete distribution [2] to efficiently optimize the latent masks.

2 Problem Setup and Notation

We formalize the environment by assuming the existence of ground-truth latent variables $\mathbf{z} = (\mathbf{z}_1, \ldots, \mathbf{z}_N)$ sampled from a distribution $p_{\mathbf{z}}$. Instead of observing $\mathbf{z}$ directly, we have access to a set of K distinct views, denoted as $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_K)$. Crucially, each view $\mathbf{x}_k$ is only a partial observation of the environment. It is generated by a specific subset of the latent variables, $\mathbf{z}_{S_k}$, where the indices are defined by a subset $S_k \subseteq [N] = \{1, \ldots, N\}$. The generation of each view is governed by a view-specific, smooth, and invertible mixing

function f_k mapping to an observation space $\mathcal{X}_k \subseteq \mathbb{R}^{\dim(\mathbf{x}_k)}$, such that $\mathbf{x}_k := f_k(\mathbf{z}_{S_k})$. See Fig. 3 for an illustration of the setup.

Let $\mathcal{V} := \{V \subseteq [K] : |V| \geq 2\}$ denote the collection of subsets $V \in \mathcal{V}$ indexing two or more views. For $V \in \mathcal{V}$, we denote the subset of views as $x_V := \{x_k : k \in V\}$. The corresponding shared content indices are given by the intersection of the view-specific indexing sets:

$$C_V = \bigcap_{k \in V} S_k$$

The associated shared latent variables are $\mathbf{z}_{C_V} := \{\mathbf{z}_j : j \in C_V\}$. Our work focuses exclusively on the global content, defined as the latent variables shared across the complete set of observed views (i.e., where $V = [K]$).

To facilitate the identification of these shared variables through learnable masks, we define a selection operator $\oslash$. For a binary mask $\mathbf{a} \in \{0,1\}^d$ and a vector $\mathbf{b} \in \mathbb{R}^d$, the operator is defined as:

$$\mathbf{a} \oslash \mathbf{b} := [\mathbf{b}_j : \mathbf{a}_j = 1, j \in [d]].$$

The primary objective of our framework—reflecting a prominent line of inquiry within multi-view causal representation learning—is to utilize the observed views to train encoders that extract all, and only, the information contained in the shared latent variables. In this context, the successful recovery of shared information is formally defined in terms of *block-identifiability.*

Definition 2.1 (Block-Identifiability)**.** *The true content variables* $\mathbf{c}$ *are block-identified by a function* $g : \mathcal{X} \to \mathbb{R}^{dim(\mathbf{c})}$ *if the inferred content partition* $\hat{\mathbf{c}} = g(\mathbf{x})$ *contains all and only information about* $\mathbf{c}$*. That is, there exists some smooth invertible mapping* h *such that* $\hat{\mathbf{c}} = h(\mathbf{c})$*.*

3 Theoretical Foundations

Our approach is inspired by the identifiability results of Yao et al. [1], which we present below. Their theoretical guarantees require the following assumptions:

Assumption 3.1 (General Assumptions)**.** *For the generative model defined in Section 2: (i) Each view-specific mixing function* f_k *is a diffeomorphism; (ii)* $p_{\mathbf{z}}$ *is a smooth and continuous density with* $p_{\mathbf{z}} > 0$ *almost everywhere.*

Their first result demonstrates that for any fixed subset $V \in \mathcal{V}$, the shared content is block-identifiable using *content encoders* g_k.

Theorem 3.1 (Identifiability from a Fixed Set of Views [1])**.** *Under Asm. 3.1, let* $G = \{g_k\}_{k \in V}$ *be a set of content encoders that minimizes:*

$$\mathcal{L}(G) = \sum_{k < k' \in V} \mathbb{E}\left[\|g_k(\mathbf{x}_k) - g_{k'}(\mathbf{x}_{k'})\|_2\right] - \sum_{k \in V} H(g_k(\mathbf{x}_k)) \tag{1}$$

where $H(\cdot)$ *denotes differential entropy and the expectation is over* $p(\mathbf{x}_V)$*. Then* $\mathbf{z}_{C_V}$ *is block-identified by* $g_k \in G$ *for any* $k \in V$*.*

To avoid retraining encoders for every subset, their second result introduces learned *content selectors* (masks) and projections to achieve simultaneous identifiability across all $V \in \mathcal{V}$ using shared *view-specific encoders.*

Theorem 3.2 (Simultaneous Identifiability via Masking [1])**.** *Under Asm. 3.1, let view-specific encoders $R = \{r_k\}$, content selectors $\Phi = \{\phi_V^k\}$, and projections $T = \{t_k\}$ solve:*

$$\min Reg(\Phi) \quad \textit{subject to:} \quad R, \Phi, T \in \arg\min \mathcal{L}(R, \Phi, T) \tag{2}$$

where $Reg(\Phi)$ is an L_0 sparsity penalty on the selectors, and the base loss is:

$$\mathcal{L}(R, \Phi, T) = \sum_{V \in \mathcal{V}} \sum_{k < k' \in V} \mathbb{E}\left[\|\phi_V^k \oslash r_k(\mathbf{x}_k) - \phi_V^{k'} \oslash r_{k'}(\mathbf{x}_{k'})\|_2\right] - \sum_{k \in V} H(t_k \circ r_k(\mathbf{x}_k)) \tag{3}$$

Then for any $V \in \mathcal{V}$ and $k \in V$, the masked representation $\phi_V^k \oslash r_k$ block-identifies $\mathbf{z}_{C_V}$.

Formal definitions and extended intuition for the encoders, selectors, and projections are deferred to Appendices A and B.

4 Method

We focus on recovering the global shared content $\mathbf{z}_{C_{[K]}}$ by aligning representations across all K views. Unlike previous theoretical frameworks that require known content or view-specific latent dimensions, we operate under a relaxed setting where all view-specific encoders r_k map to a unified latent space of dimension d. We assume only that this common dimension d is an overestimation of the true content dimension.

To bridge the gap between theory and practice, we introduce two primary modifications. First, we implement a *shared masking mechanism* where a single binary mask $\phi \in \{0, 1\}^d$ is applied globally across the outputs of all encoders $\{r_k\}_{k=1}^K$. This enforces a common latent subspace for shared content and significantly simplifies the per-view selectors required by Theorem 3.2. We optimize this mask end-to-end using a *hard concrete distribution* [2], which enables differentiable L_0 regularization while yielding exact binary values to prune unshared dimensions.

Second, we replace the intractable differential entropy terms in the theoretical objective with a *Symmetric InfoNCE* loss [3, 4]. This self-supervised surrogate enforces alignment through pairwise contrast across all views while preventing dimensional collapse by maximizing a lower bound on mutual information. The total objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SymInfoNCE}} + \lambda \text{Reg}(\phi). \tag{4}$$

Detailed formulations of the masking distribution and the contrastive objective are provided in Appendix C.

5 Experiments

Our experiments follow the same flavor as in [5] and [1]: we first evaluate the performance of our method on synthetic data, where we have access to the ground truth latent variables. Then, we apply our method to the *Multimodal3DIdent* dataset [5], which consists of blender-rendered images of 3D objects, paired with descriptive text captions. The code is available in our Github repository[1] and builds upon the code of Daunhawer et al. [5] and Yao et al. [1].

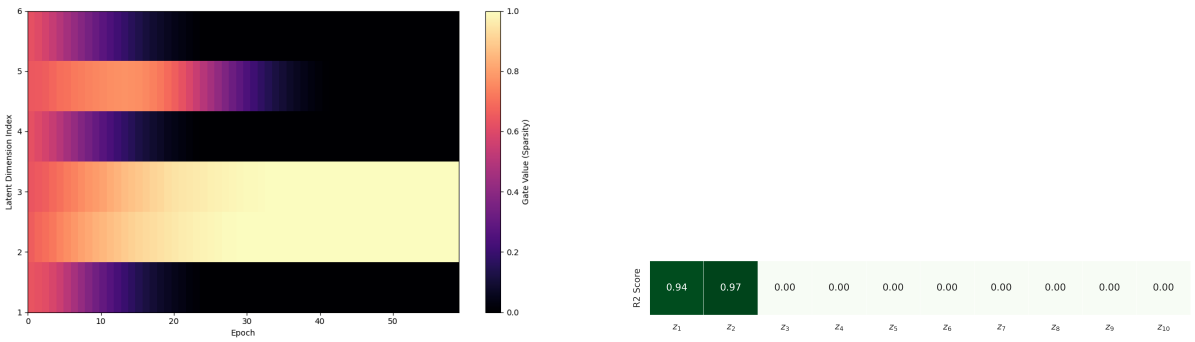
[1] `https://github.com/Virusmint/multiview-sparse`

To evaluate block-identifiability, we assess whether the learned representations capture all and only the information contained in the shared content. We quantify this by training downstream predictors to recover the ground-truth latent factors directly from the learned embeddings. Successful block identification is characterized by high predictive performance for shared content variables and near-chance performance for style variables, except in cases where significant statistical dependencies exist. Specifically, we report the R^2 scores from linear regression for continuous latent factors and the classification accuracy from multinomial logistic regression for discrete factors.

5.1 Numerical Experiment

We generate synthetic data in which 10 *independent* latent variables are sampled from a Gaussian distribution $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I}_\mathbf{z})$. Furthermore, we create 8 views from the latent variables, where each view $\mathbf{x}_m$ is generated by an invertible nonlinear transformation of the latent variables, $\mathbf{x}_m = f_m(\mathbf{z}_{S_m})$, where S_m is the subset of latent variables that underlie view m. Importantly, only 2 latents $\mathbf{z}_1$ and $\mathbf{z}_2$ are shared (content) across all views, while the remaining latents are view-specific. Moreover, the mixing functions f_m are implemented as 2-layer *invertible, untrainable* MLPs with LeakyRELU - invertibility is guaranteed by applying an orthogonal initialization to the weights. The view encoders are implemented as 2-layer *trainable* MLPs, each with an overestimated output dimension of 6. We summarize our results in Fig. 1 and defer the implementation details to Tab. 1a.

We also defer the results of an additional experiment to Appendix D.3, where *causal* dependencies are introduced between the latent variables.



(a) Gate convergence history.

(b) R^2 scores per dimension.

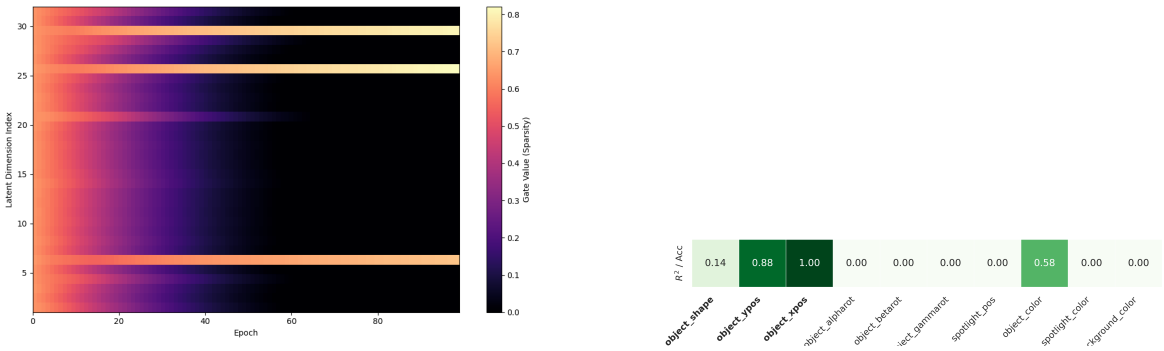
Figure 1: Numerical experiment results: (a) convergence of the hard concrete gate across the six latent dimensions over the training duration. The gate maintains exactly two active dimensions in the end, which is consistent with the true number of content factors; (b) the R^2 scores show that the content factors are successfully block-identified, whereas style factors ignored.

5.2 Multimodal Experiment

We evaluate our approach on the Multimodal3DIdent dataset [5], which pairs 3D image renderings (10 latent factors) with descriptive captions (5 latent factors). Crucially, only three factors—*object shape*, *x-position*, and *y-position*—are shared across modalities,

constituting the global shared content $\mathbf{z}_{C_{[K]}}$ to be recovered. While both modalities include an "object color" latent, they are treated as style variables because they lack a bijective mapping; the image utilizes a continuous hue value, whereas the caption uses a discrete color name. Our results are presented in Fig. 2

Detailed latent specifications are provided in Appendix D.1, along with implementation details in Tab. 1b.



(a) Gate convergence history.

(b) Prediction performance per latent factor.

Figure 2: Multimodal experiment results: (a) convergence of the hard concrete gate across 32 latent dimensions over the training duration; exactly three dimensions are kept active in the end, which is consistent with the true number of content factors. (b) the prediction performance is similar to [5]'s results; consistent with their findings, the recovery of the content `object_shape` proved difficult, while the style `object_colour` factor achieved a high R^2 score due to the strong dependency between the different colour latents specific to the image and text views.

6 Limitations and Future Work

While our empirical results are strong, this work lacks formal proofs for the relaxed setting in which the only assumption is the overestimation of the content dimension. Future research should establish the theoretical conditions required for block-identifiability under these assumptions. Additionally, the impact of the degree of overestimation on recovery accuracy and training cost remains to be explored. Investigating the consequences of underestimation would also be an avenue of practical relevance for our approach.

7 Conclusion

In this work, we addressed the challenge of bridging the gap between theoretical identifiability and practical implementation in multi-view representation learning. Specifically, we introduced a differentiable shared masking mechanism that relaxes the restrictive requirement for prior knowledge of the latent space's exact dimensionality. By allowing for an overestimation of the content dimension, our approach provides a more robust and realistic framework for recovering global shared content. Empirical evaluations demonstrate that our method achieves performance comparable to previous works [5, 1] that rely on oracle knowledge of latent dimensionality.

Self-Assessment

We feel quite satisfied with the outcome of the project, as we have not only addressed an important limitation of the existing literature (i.e., the assumption of known content dimension), but also proposed a practical solution that is both computationally efficient and elegant in identifying the global content. Given a second iteration, we would like to investigate the theoretical properties of our method to understand why it works so well in practice.

Statement on the Use of LLMs

LLMs were used mostly for the coding portion of the project - partly to understand pre-written codebases from Yao and Daunhawer, and partly to assist in the implementation of our methods. They were also used in the writing portion to accelerate the drafting/formatting of the report (which was then thoroughly edited by us). LLMs were also used as a research tool to aid in finding and understanding relevant literature.

Statement on Contributions

David wrote the majority of the code for the model architectures and designed the numerical experiment. He also handled the writing about the experiment implementations and results and contributed significantly to the editing of the report. Ryan wrote and edited the majority of the report and established most of the code for the multimodal experiment. Dowoo prepared part of the final presentation.

Acknowledgements

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[5] Imant Daunhawer, Alice Bizeul, Emanuele Palumbo, Alexander Marx, and Julia E. Vogt. Identifiability results for multimodal contrastive learning. 2023.

Appendix

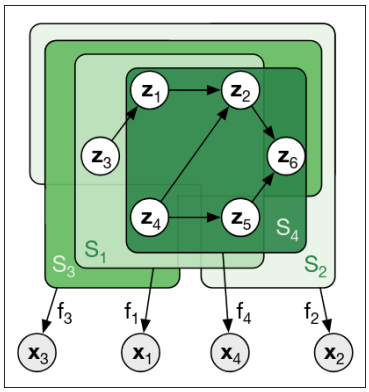


Figure 3: Example of a multi-view setup under partial observability. Source: [1]

A Definitions for Section 3

Definition A.1 (Content Encoders)**.** *Assuming the content size $|C|$ is given for a set of jointly observed views x_V, the content encoders $G := \{g_k : \mathcal{X}_k \to (0,1)^{|C|}\}_{k \in V}$ consist of smooth functions mapping the respective observation spaces to the $|C|$-dimensional unit cube.*

Definition A.2 (View-Specific Encoders)**.** *The view-specific encoders $R := \{r_k : \mathcal{X}_k \to \mathcal{Z}_{S_k}\}_{k \in V}$ consist of smooth functions mapping from the respective observation spaces to the view-specific latent space, where the dimension of the k-th latent space $|S_k|$ is assumed to be known.*

Definition A.3 (Content Selectors)**.** *The content selectors $\Phi := \{\phi_V^k : V \in \mathcal{V}, k \in V\}$ are learnable binary vectors, with $\phi_V^k \in \{0,1\}^{|S_k|}$, that perform element-wise selection (denoted by $\oslash$) on the encoded representation such that $\phi_V^k \oslash r_k(x_k)$ extracts the selected representation.*

Definition A.4 (Projections)**.** *The set of projections $T := \{t_k\}_{k \in V}$ consist of smooth functions $t_k : \mathcal{Z}_{S_k} \to (0,1)^{|S_k|}$ mapping each view-specific latent space to a hyper unit-cube of the same dimension $|S_k|$. These enforce invertibility of the encoders without interfering with disentanglement.*

B Intuition for Mathematical Components

B.1 Content Selectors (Φ)

The content selectors act as a "soft index" or a routing mechanism. Since the encoder r_k does not know a priori which dimensions of its output will match the dimensions of another view's encoder, the selectors ϕ learn to point to the correct coordinates. The information-sharing regularizer $\text{Reg}(\Phi)$ acts as an Occam's Razor: it pushes the model to explain the correlation between views using the fewest number of latent dimensions possible. Without this regularizer, the model might trivially designate all dimensions as "shared", failing to distinguish between universal content and view-specific style.

B.2 Projections (T)

The projection functions t_k are introduced to resolve a conflict between invertibility and causal dependency. Standard contrastive learning often forces representations to be independent and uniform (to maximize entropy). However, if the true underlying causal variables are dependent, they cannot be independent.

The projection t_k acts as a buffer. By maximizing the entropy of the *projected* representation $t_k(r_k(x_k))$, the framework ensures the mapping r_k is invertible (preserving all information) without forcing the latent variables themselves to be independent. This allows the model to correctly identify variables that have causal links to one another while still satisfying the mathematical requirements for a non-degenerate solution.

C Detailed Approach and Formulations

C.1 Hard Concrete Masking

The discrete L_0 penalty is non-differentiable. Following [2], we treat each entry of the shared mask ϕ_j as a stochastic gate sampled from a Bernoulli distribution. To enable gradient-based optimization, we use the hard concrete distribution to "stretch" a continuous variable s into the interval (γ, ζ), where $\gamma < 0$ and $\zeta > 1$. The mask $\tilde{\phi}_j$ is obtained via a hard-sigmoid:

$$s = \text{Sigmoid}\left(\frac{\log u - \log(1-u) + \log \alpha_j}{\beta}\right), \quad \tilde{\phi}_j = \min(1, \max(0, s(\zeta - \gamma) + \gamma)) \tag{5}$$

where $u \sim \mathcal{U}(0,1)$, $\log \alpha_j$ is the location parameter, and β is the temperature. This allows the information-sharing regularizer $\text{Reg}(\phi)$ to be implemented as a fully differentiable complexity loss:

$$\text{Reg}(\phi) = \sum_{j=1}^{d} \text{Sigmoid}\left(\log \alpha_j - \beta \log \frac{-\gamma}{\zeta}\right). \tag{6}$$

At test time, we obtain deterministic masks using the expected estimator:

$$\hat{\phi}_j = \min(1, \max(0, \text{Sigmoid}(\log \alpha_j)(\zeta - \gamma) + \gamma)). \tag{7}$$

C.2 Multiview Symmetric InfoNCE

Let $\hat{z}_k = \tilde{\phi} \oslash r_k(x_k)$ be the masked representation for view k. To ensure joint alignment across all K views, we employ a Symmetric InfoNCE loss [3, 4], aggregated across all unique pairs (k, k'):

$$\mathcal{L}_{\text{SymInfoNCE}} = \frac{1}{K(K-1)B} \sum_{k=1}^{K} \sum_{k' \neq k} \sum_{i=1}^{B} -\log \frac{e^{\text{sim}(\hat{z}_{i,k}, \hat{z}_{i,k'})/\tau}}{\sum_{j=1}^{B} e^{\text{sim}(\hat{z}_{i,k}, \hat{z}_{j,k'})/\tau}} \tag{8}$$

where B is the batch size, $\text{sim}(\cdot,\cdot)$ denotes the similarity measure (e.g., negative Euclidean distance or cosine similarity) and τ is the temperature parameter. The denominator acts as a surrogate for entropy maximization by contrasting paired representations against batch distractors, preventing the trivial solution of dimensional collapse.

D Experiments

D.1 Experimental Details

Multimodal Experiment with Multimodal3DIdent The Multimodal3DIdent dataset [5] consists of $224 \times 224 \times 3$ RGB images paired with descriptive natural language captions. The images are rendered in Blender using 10 underlying factors: object shape (7 classes), x and y positions (3 discrete values each), 3D rotations (3 continuous angles), object colour, spotlight position, spotlight colour, and background colour. The corresponding text is generated from 5 factors: shape, x-position, y-position, colour names selected from random palettes, and one of five unique sentence templates that determine the phrasing style. Examples of the dataset are displayed in Fig. 4

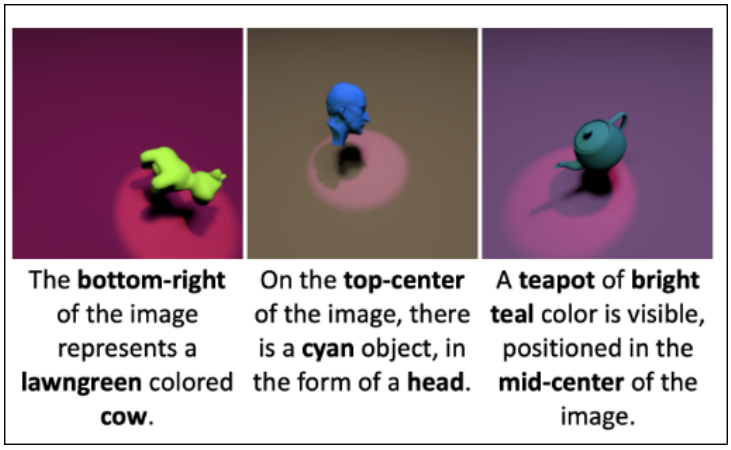


Figure 4: Multimodal3DIdent Dataset - examples of image-caption pairs. Source: [5]

The generative process is designed so that only three factors—object shape, x-position, and y-position—are identical across both modalities. While both views reference the object's colour, the image uses continuous hue values while the text uses discrete names from varying palettes, preventing a direct one-to-one correspondence. All other factors, such as camera rotation, lighting, and sentence structure, are modality-specific.

D.2 Implementation Details

In Tab. 1, we lay out the model architectures and hyperparameters used for both the numerical and multimodal studies.

D.3 Additional Numerical Experiment Results

We remodel the numerical study to allow for *causally dependent* latent factors $\mathbf{z} \sim \mathcal{N}(0, \mathbf{\Sigma_z})$ via a non-trivial covariance matrix $\Sigma_\mathbf{z} \sim \text{Wishart}(0, \mathbf{I_z})$. Even under this formulation, Tab. 5 shows that the content factors are block-identified using the correct number of dimensions. This shows that our method is also robust to causal dependencies between the latent variables, which is a common scenario in real-world applications.

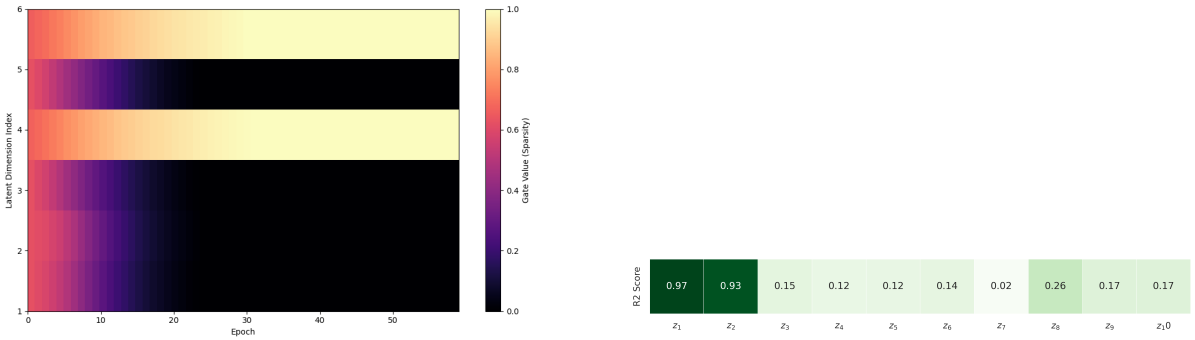
Parameter	Value
Generating function	2-layer invertible MLP
Encoder	2-layer MLP
Output dim. d	6
Optimizer	Adam
Cond. threshold ratio	1e-3
Batch size	4096
Learning rate	1e-3
Temperature τ	2.0
# Iterations	6,000
Similarity metric	Cosine similarity

(a) Numerical simulation.

Parameter	Value
Generating function	Image and text rendering
Image encoder	ResNet-18
Text encoder	4-layer ConvNet
Output dim. d	32
Optimizer	Adam
Batch size	256
Learning rate	1e-4
Temperature τ	2.0
# Iterations	50,000
# Samples (train/val/test)	125k/10k/10k
Similarity metric	Cosine similarity

(b) *Multimodal3DIdent*.

Table 1: Experiment Hyperparameters. Note that our experiments only use a fraction of the number of iterations used in [1] and [5], for computational efficiency, though the results remain competitive enough for comparison.



(a) Gate convergence history.

(b) R^2 scores per latent factor.

Figure 5: Numerical experiment with dependent factors: (a) even in the case where the latent factors are causally dependent, the model still learns the correct number of content dimensions - (b) the R^2 scores also indicate that block-identifiably has been achieved, and the style factors now have non-zero R^2 values due to the latent dependencies.